TASK 4 — STEP-BY-STEP GUIDE

DESIGN AN INTERACTIVE iOS PROTOTYPE WITH FIGMA

Course: Geo Programming | Week 5 | Suggested time: 45–60 minutes | Individual work

1. Learning Outcomes

Identify justified screens; build a clickable Figma prototype; connect all necessary navigation paths; audit every use case and functional requirement against visible UI actions and feedback; plan SwiftUI View/controller responsibilities.

2. Scenario & Inputs

School Management System (2026–2027): staff browse students, subjects, teachers and enrollments; add/edit students; assign subjects; enroll students; enter/update marks from 0 to 10; and inspect results. Duplicate enrollments must be rejected.

Use the functional requirements (Task 1), use cases (Task 2) and OOP models (Task 3). Design and prototype first; do not implement the full SwiftUI app in this task.

Step 1 — Identify Screens and Their Purposes

Derive screens from user goals, not one screen per class. Three worked examples:

Screen ID / name	Purpose	Related use case
S01 Student List	Browse student records	UC: Browse Students
S02 Student Detail	Inspect student, subjects and marks	UC: Inspect Student Results
S03 Student Form	Add or edit student	UC: Add/Edit Student

YOUR TURN — Identify ALL remaining justified screens

List every screen needed for remaining use cases (subjects, teachers, enrollment, marks, results), including error/empty states when distinct designs are necessary. Add rows as needed.

Screen ID / name	Purpose	UC ID(s)
S04 Subject List	Browse subjects (ID, title, credits, semester) and who teaches them	UC-08
S05 Teacher List	Browse teachers and number of subjects taught; entry to assignment	UC-08, UC-06
S06 Teacher – Assign Subjects	Toggle the subjects a teacher teaches and save	UC-06
S07 Enrollment List	Browse all enrollments (student, subject, year, mark); entry to enroll / mark	UC-08, UC-03
S08 Enrollment Form	Choose student, subject and academic year, then enroll	UC-02
S09 Enrollment / Mark Entry	View one enrollment and enter or update the mark	UC-03
S10 Student List – Empty / No results	Empty-state design for no students or no search match	UC-01
S11 Student Form – Validation Errors	Inline errors for blank ID/name, invalid email, duplicate ID	UC-04, UC-05 (error flows)

S12 Enrollment – Duplicate Error	Error banner + field error; nothing is saved	UC-02 (error flow)
S13 Mark – Invalid Value	Inline “Mark must be 0–10”; saved mark unchanged	UC-03 (error flow)
S14 Student Detail – No Enrollments	Empty state for a student without enrollments	UC-07
S15 Student List – Saved	Success feedback after adding a student (new row highlighted)	UC-04
S16 Student Detail – Saved	Success feedback after enrolling or saving a mark (row highlighted)	UC-02, UC-03

Steps 2–4 — Design Screens and Connect Them in Figma

These three former paper-sketch steps are replaced by ONE practical Figma workflow. Use https://www.figma.com/ and create a design file named GeoProgramming_Task4_StudentID.

A. Create the design file and screen frames

1. Create a new Figma Design file. Add a page named Screens.
2. Press F (Frame) and select a consistent iPhone-sized preset. Name each frame with its screen ID and purpose.
3. Use shapes, text and reusable components to create the screen layout. Include data labels, input fields, actions and user feedback.
4. Create all justified screens identified in Step 1, not just the three examples.

Three guided screen examples

Frame	Visible elements	Actions / feedback
S01 Student List	Title, search field, student rows, Add (+)	Tap row; tap Add; empty-list message
S02 Student Detail	Student ID/name, enrolled subjects, marks, Edit	Tap Edit; navigate back
S03 Student Form	ID, name, email, Save, Cancel	Save success; required-field error

B. Connect screens with Figma Prototype

1. Switch to the Prototype tab. Select an interactive element, then drag its prototype connector to the destination frame.
2. Set trigger On click / On tap and action Navigate to (or Open overlay for a modal).
3. Define a starting frame. Add Back, Cancel and Save success paths.
4. Use separate frames or overlays to demonstrate invalid input, duplicate enrollment and other feedback.
5. Click Present / Play and test the prototype end to end.

Three guided prototype connections

From	Tap / trigger	To / outcome
S01 Student List	Tap student row	S02 Student Detail
S01 Student List	Tap Add (+)	S03 Student Form
S02 Student Detail	Tap Edit	S03 Student Form (edit state)

YOUR TURN — Build and connect ALL remaining screens

Create the rest of the screens and interactions in Figma. Test forward and return routes; include success, cancellation, empty data and validation feedback. Record unresolved links below.

Source frame	Element / action	Destination / state	Tested?

S03 Student Form	Tap Cancel	S01 (or S02 when editing) – nothing saved	☐ test in Present mode
S03 Student Form	Tap Save (valid input, new student)	S15 – "Student saved", new row highlighted	☐ test in Present mode
S03 Student Form (edit)	Tap Save (valid)	S02 – updated name/email shown	☐ test in Present mode
S03 Student Form	Tap Save with blank ID / invalid email	S11 – inline errors, no record created	☐ test in Present mode
S11 Student Form – Errors	Fix fields and tap Save	S15	☐ test in Present mode
S01 Student List	Type "zzz" in search	S10 – "No students found"	☐ test in Present mode
S02 Student Detail	Tap Enroll (+ Enroll)	S08 with the student preselected	☐ test in Present mode
S02 Student Detail	Tap an enrollment row	S09 Enrollment / Mark Entry	☐ test in Present mode
S02 Student Detail (no enrollments)	Tap "Enroll in a subject"	S08 (S14 is the empty variant of S02)	☐ test in Present mode
S08 Enrollment Form	Tap Enroll – new combination	S16 – "Enrollment saved", new row highlighted	☐ test in Present mode
S08 Enrollment Form	Tap Enroll – duplicate combination	S12 – duplicate error; tap field to correct → S08	☐ test in Present mode
S08 Enrollment Form	Tap Cancel	Previous screen (S02 or S07)	☐ test in Present mode
S09 Mark Entry	Enter 8.5 → Save	S16 (text "Mark saved") – updated mark visible	☐ test in Present mode
S09 Mark Entry	Enter 11 → Save	S13 – inline "Mark must be 0–10"; edit field → S09	☐ test in Present mode
S09 Mark Entry	Tap ‹ Student	S02	☐ test in Present mode
Tab bar (all main screens)	Tap Students / Subjects / Teachers / Enrollments	S01 / S04 / S05 / S07	☐ test in Present mode
S05 Teacher List	Tap a teacher row	S06 Teacher – Assign Subjects	☐ test in Present mode
S06 Assign Subjects	Toggle subjects → Save assignments (or ‹ Teachers to cancel)	S05 – subject count updated / unchanged	☐ test in Present mode
S07 Enrollment List	Tap + / tap a row	S08 / S09	☐ test in Present mode
S15 / S16	Tap a row / ‹ Students	S02 / S01	☐ test in Present mode

Step 5 — Audit Use Cases, Functions and UI Mapping

Goal: prove that EVERY functional requirement (FR) from Task 1 and EVERY use case (UC) from Task 2 is represented in the Figma prototype. Do not treat a screen title alone as proof: identify the exact control, action and observable feedback.

A. Make a master inventory

1. Copy ALL FR IDs and short descriptions from Task 1.
2. Copy ALL UC IDs from Task 2, including alternate/error flows.
3. For each FR, identify the related UC(s). For each UC, identify the FR(s) it satisfies. Mark missing links as gaps, not as completed.

B. Trace each requirement to the interface

4. Identify the Figma frame ID, exact UI control, user action, destination/state and expected feedback.
5. Check the prototype in Present mode. Mark Pass only if the intended interaction and feedback are actually represented.
6. Record missing functionality, missing screens and unclear error states; revise Figma and recheck.

Three worked mapping examples (illustrative IDs; use your actual Task 1–2 IDs)

FR / UC	Figma frame + UI action	Success / error feedback	Status
Browse students / Browse Students	S01 → open Students list	Rows displayed / "No students yet"	Example
Edit student / Add/Edit Student	S03 → enter values → Save	Updated detail / invalid email message	Example
Update mark / Update Mark	Enrollment Detail → enter mark → Save	Updated mark / "Mark must be 0–10"	Example

YOUR TURN — Complete the FULL traceability matrix

Enter EVERY remaining FR and UC, including alternate flows. Extend the table until there are no uncovered requirements.

FR ID	UC ID / flow	Frame ID	UI control + action	Feedback / destination	Pass / gap
FR-01	UC-01	S01	Students tab → rows show ID + name; search field	Rows displayed; S10 "No students found" when empty	Pass (wireframe)
FR-01 (empty)	UC-01 alt	S10	Search "zzz" / no data	"No students found / No students yet" + Add student button	Pass (wireframe)
FR-02	UC-04	S01 → S03 → S15	Tap + → enter ID, name, email → Save	S15 green "Student saved", new row highlighted	Pass (wireframe)
FR-03	UC-05	S02 → S03 → S02	Tap Edit → change name/email → Save	S02 shows updated values	Pass (wireframe)
FR-04	UC-06	S05 → S06 → S05	Tap teacher → toggle subjects → Save assignments	S05 count updated; S04 lists the teachers per subject	Pass (wireframe)
FR-05	UC-02	S02 / S07 → S08 → S16	Tap + Enroll → pick student, subject, year → Enroll	S16 "Enrollment saved", new row highlighted	Pass (wireframe)
FR-06	UC-02 alt: duplicate	S08 → S12	Enroll the same student+subject +year again	Red banner + field error "Already enrolled"; nothing saved	Pass (wireframe)
FR-07	UC-03	S02 → S09 →	Tap enrollment	Updated mark	Pass

		S16	row → enter 8.5 → Save	shown; “Mark saved”	(wireframe)
FR-07 (invalid)	UC-03 alt: mark = 11	S09 → S13	Enter 11 → Save	Inline “Mark must be 0–10”; old mark kept; no success banner	Pass (wireframe)
FR-08	UC-07	S01 → S02	Tap a student row	Subjects, academic years and marks listed	Pass (wireframe)
FR-08 (empty)	UC-07 alt	S14	Open a student without enrollments	“No enrollments yet” + Enroll button	Pass (wireframe)
FR-09	UC-04 / UC-05 alt: required field blank	S03 → S11	Save with blank ID / name	Inline “Student ID is required”; no record created	Pass (wireframe)
FR-09, FR-12	UC-04 alt: invalid email / duplicate ID	S11	Save with “em.nguyen@school”	Inline “Enter a valid email…”; variant text “ID already exists”	Pass (wireframe)
FR-10 (proposed)	UC-08	S04, S05	Subjects / Teachers tabs	Subject rows (ID, title, credits, semester, teachers); teacher rows	Pass (wireframe); FR missing in Task 1 → Gap 1
FR-11 (proposed)	UC-08	S07	Enrollments tab	Rows: student, subject, year, mark or “—”	Pass (wireframe); FR missing in Task 1 → Gap 1
FR-12 (proposed)	UC-04, 05, 02	S11, S08	Email text field; academic-year picker (prevents bad format)	Inline format message; picker only offers valid years	Pass (wireframe)
Cancel / back	All UCs	S03, S08, S06, S09	Cancel / ‹ Back	Return to the previous screen without saving	Pass (wireframe)

Step 5 — Checklist for Functional Coverage (continued)

Use this checklist AFTER testing the Figma prototype. A requirement is covered only when the screen, action and feedback can be demonstrated.

☐ Every FR from Task 1 appears in the traceability matrix.

☐ Every UC from Task 2 appears, including alternate and error flows.

☐ Each mapped item names a specific Figma frame and interactive control.

☐ Every key action has a visible success result or destination.

☐ Invalid marks (outside 0–10) show a clear error and do not imply a successful save.

☐ Duplicate enrollment produces an error state and does not imply a duplicate record was created.

☐ Required fields and other documented validation rules have feedback.

☐ Empty data and cancel/back navigation are demonstrated where relevant.

☐ All Figma prototype connections have been tested using Present mode.

☐ Any unsupported requirement is marked GAP and has a revision plan.

Gap log — identify and fix ALL uncovered items

FR / UC	What is missing?	Figma change required	Retest result
FR-10, FR-11 / UC-08	Task 1 has no requirement for browsing subjects, teachers or enrollments, yet the scenario needs them	Added S04, S05, S07 and the tab bar; add FR-10/FR-11 to Task 1	☐ Recheck in Present mode
FR-12 / UC-04, 05	Email, credits and academic-year format rules are not stated as an FR	Inline messages in S11; year chosen from a picker in S08	☐ Recheck
UC-04 alt	Duplicate student ID had no UI state	S11 variant text “Student ID already exists”	☐ Recheck
UC-02, UC-03	Success feedback missing after save	Added S15 and S16 with green banners and highlighted row	☐ Recheck
UC-01, UC-07	Empty states were not designed	Added S10 and S14	☐ Recheck
S01 search field	Search is not required by any FR	Keep as optional convenience or remove; decide before coding	☐ Decide
NFR-02, NFR-03	Layout and error placement must meet iPhone / “beneath the field” criteria	All frames are 390×844 portrait; errors sit directly under the field	☐ Check in Present mode
All connections	Figma file/prototype not created here	Import the 16 SVG frames, wire the connections above, test	☐ Pending

Student reflection

Which FR or UC was hardest to map to an interface action? Explain your design decision in 3–5 sentences.

The hardest requirement to map was the duplicate-enrollment rule (FR-06), because it is not an action the user performs but a rule that blocks one. I showed it as a distinct error state (S12) with a red banner and a field-level message, and no success banner, so the prototype never suggests that a second record was created. The mark rule (FR-07) was similar: S13 shows the error directly beneath the field and keeps the saved mark unchanged. While auditing I also found that browsing subjects, teachers and enrollments had no functional requirement, so I recorded it in the gap log and proposed FR-10 and FR-11.

Step 6 — Plan SwiftUI Views and Controller Responsibilities

Use the Figma design as a guide for later SwiftUI implementation. UI elements display data and collect actions; validation and business rules belong in appropriate model/service/controller logic.

Three worked examples

Proposed Swift file / component	Responsibility	Business rule / service
StudentListView.swift	Render student rows and navigation	StudentController.loadStudents()

StudentFormView.swift	Collect input; show errors	StudentController.saveStudent(...)
EnrollmentView.swift	Choose student/subject; show result	EnrollmentService.enroll(...) with duplicate check

YOUR TURN — Identify ALL remaining views and coordinating components

Figma frame	SwiftUI View file	Controller / service	Validation / state owner
S01, S10, S15	StudentListView.swift (+ empty-state subview)	StudentController.loadStudents(matching:)	View owns search text; StudentController owns the data
S02, S14, S16	StudentDetailView.swift	EnrollmentService.results (forStudent:)	View shows rows/empty state; service builds the results
S03, S11	StudentFormView.swift	StudentController.addStudent / updateStudent	Student.init validates; form keeps @State error text per field
S04	SubjectListView.swift	TeachingAssignmentService.teachers(forSubject:)	Read-only; no rules
S05	TeacherListView.swift	SchoolRepository (via controller)	Read-only; no rules
S06	TeacherAssignmentView.swift	TeachingAssignmentService.assign / unassign	Teacher.assign rejects duplicates; view keeps toggle state
S07	EnrollmentListView.swift	EnrollmentService (list of enrollments)	Read-only; no rules
S08, S12	EnrollmentFormView.swift	EnrollmentService.enroll(...)	Duplicate check in EnrollmentService; view shows the ValidationError banner
S09, S13	MarkEntryView.swift	EnrollmentService.updateMark(...)	Enrollment.updateMark enforces 0–10; view parses text and shows the error
All	ContentView.swift (TabView) + ToastView / ErrorBannerView.swift	SchoolRepository injected once via the environment	No logic; banners display the message mapped from ValidationError

Step 7 — Review, Test and Submit

Open the Figma prototype in Present mode and test every mapped FR/UC. Confirm screens, navigation, Save/Cancel paths and validation feedback. Fix all recorded gaps before submission.

Three worked review examples

Review item	Test	Expected result
Coverage	Open Student List from entry screen	Student rows / empty state visible
Navigation	Open Student Form, then Cancel	Return without saving
Validation	Try enrolling the same student twice	Duplicate warning state visible

YOUR TURN — Record ALL remaining checks

FR / UC / screen	Test action	Pass / gap	Revision
FR-01 / S01	Open the Students tab	Pass (wireframe)	—

FR-01 / S10	Type “zzz” in search	Pass (wireframe)	—
FR-02, FR-09 / S03, S11	Save with blank ID and bad email	Pass (wireframe)	Added S11
FR-03 / S03	Edit a student, Cancel	Pass (wireframe)	—
FR-04 / S06	Toggle two subjects, Save	Pass (wireframe)	—
FR-05, FR-06 / S08, S12	Enroll twice in the same subject and year	Pass (wireframe)	Added S12
FR-07 / S09, S13	Enter 8.5, then 11	Pass (wireframe)	Added S13
FR-08 / S02, S14	Open student with and without enrollments	Pass (wireframe)	Added S14
FR-10, FR-11 / S04, S05, S07	Use the tab bar	Pass (wireframe); FRs to be added to Task 1	Added S04, S05, S07
Whole prototype in Figma	Play from S01 and follow every connection	GAP – pending (Figma not built yet)	Import SVGs, connect, test

Required Deliverables

☐ Shareable Figma design file link (view access enabled for instructor).

☐ Clickable Figma prototype with complete navigation and feedback states.

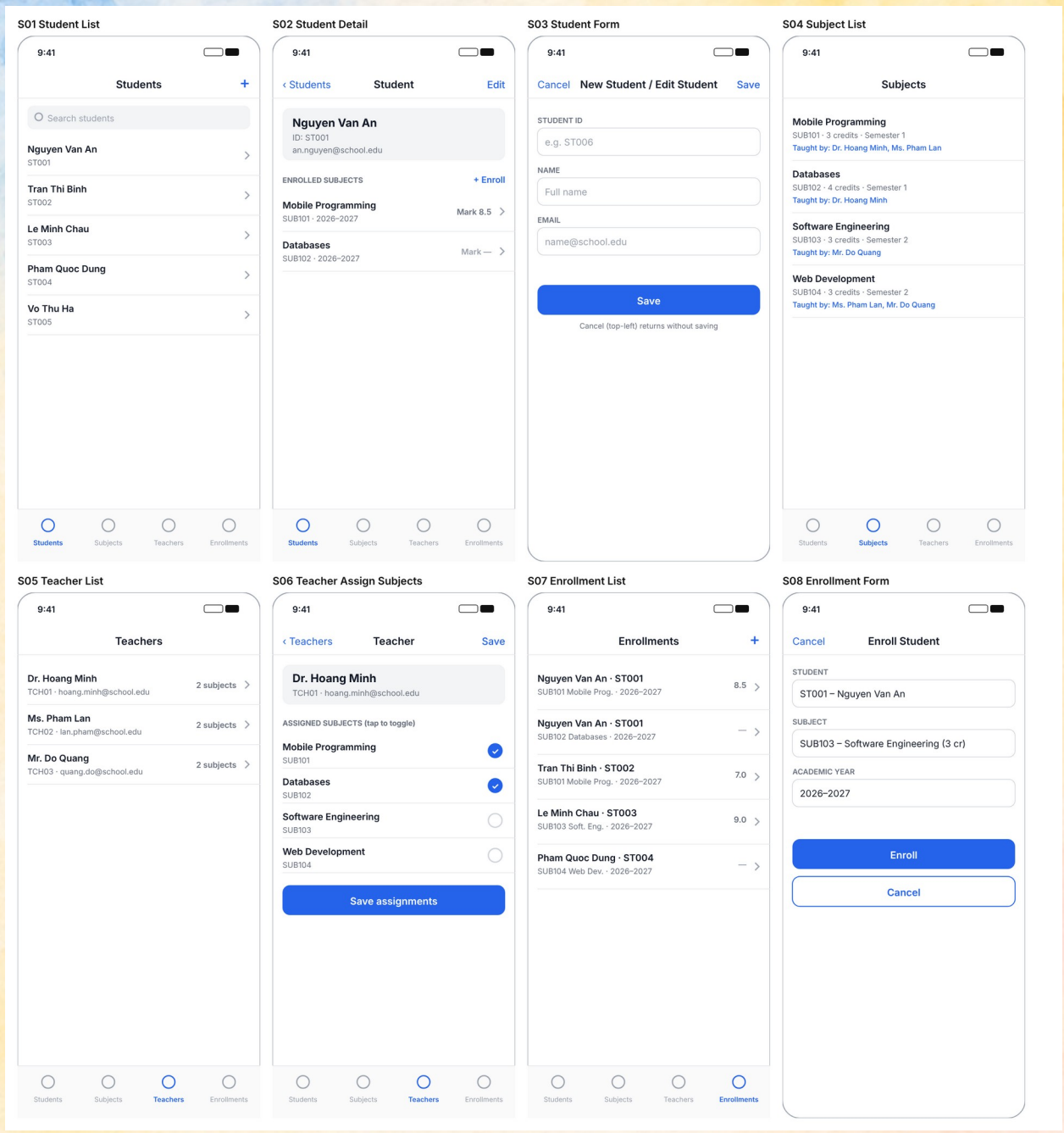
☐ Screen inventory with related UC IDs.

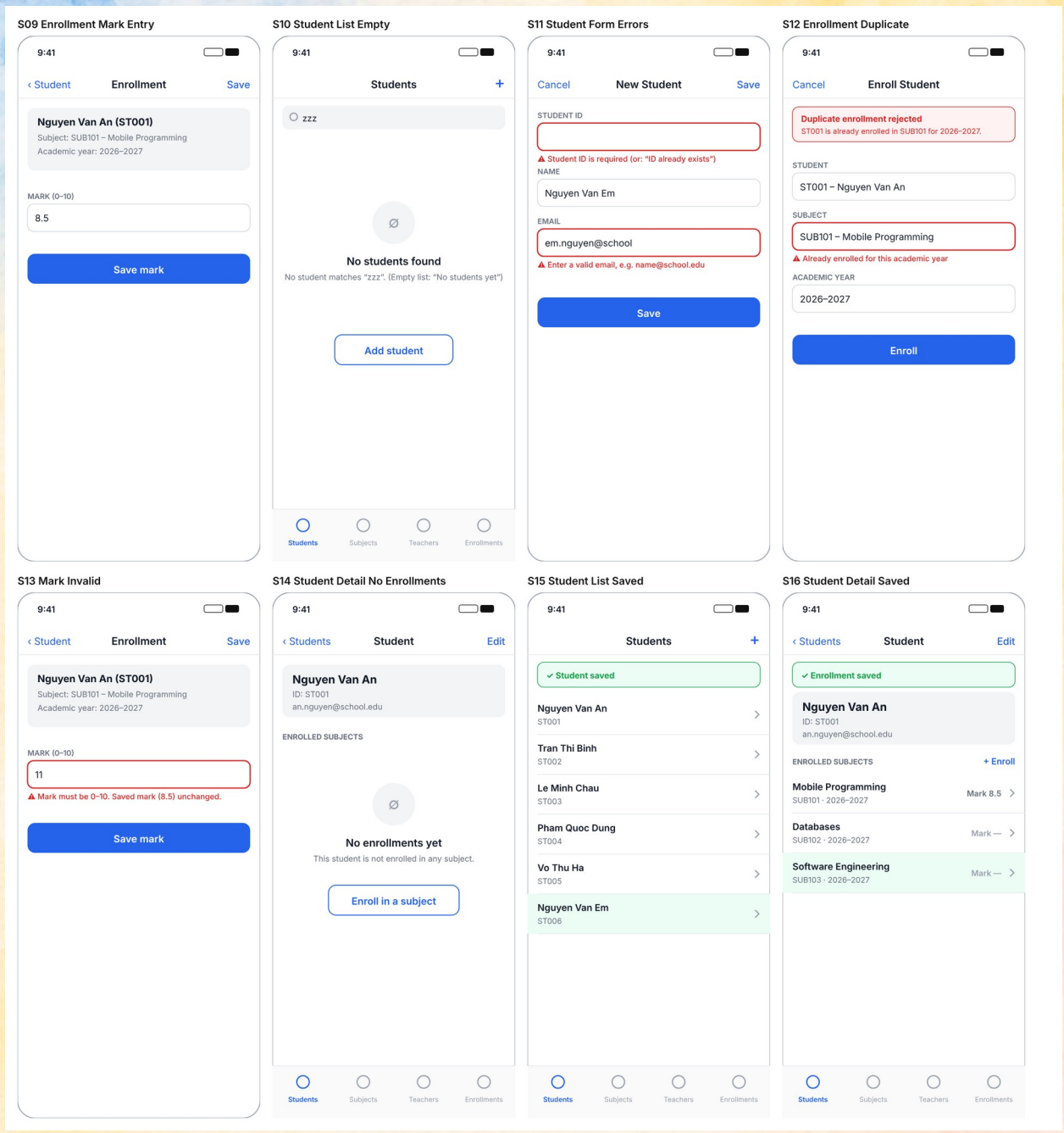
☐ Completed FR–UC–screen–action–feedback traceability matrix and checklist.

☐ Gap log and proposed SwiftUI View/controller structure.

Submit the Figma link and export your traceability worksheet as Task4_Figma_StudentID.pdf with Lab 01.

Appendix – Wireframe screens S01–S16 (portrait iPhone, 390×844). Each screen is also supplied as a separate .svg file that can be dragged into Figma as an editable frame.


S01 Student List
9:41
Students
Search students
Nguyen Van An
ST001
Tran Thi Binh
ST002
Le Minh Chau
ST003
Pham Quoc Dung
ST004
Vo Thu Ha
ST005
Students
Subjects
Teachers
Enrollments
S02 Student Detail
9:41
‹ Students
Student
Edit
Nguyen Van An
ID: ST001
an.nguyen@school.edu
ENROLLED SUBJECTS
+ Enroll
Mobile Programming
SUB101 · 2026–2027
Mark 8.5
Databases
SUB102 · 2026–2027
Mark —
Students
Subjects
Teachers
Enrollments
S03 Student Form
9:41
Cancel
New Student / Edit Student
Save
STUDENT ID
e.g. ST006
NAME
Full name
EMAIL
name@school.edu
Save
Cancel (top-left) returns without saving
S04 Subject List
9:41
Subjects
Mobile Programming
SUB101 · 3 credits · Semester 1
Taught by: Dr. Hoang Minh, Ms. Pham Lan
Databases
SUB102 · 4 credits · Semester 1
Taught by: Dr. Hoang Minh
Software Engineering
SUB103 · 3 credits · Semester 2
Taught by: Mr. Do Quang
Web Development
SUB104 · 3 credits · Semester 2
Taught by: Ms. Pham Lan, Mr. Do Quang
Students
Subjects
Teachers
Enrollments
S05 Teacher List
9:41
Teachers
Dr. Hoang Minh
TCH01 · hoang.minh@school.edu
2 subjects
Ms. Pham Lan
TCH02 · lan.pham@school.edu
2 subjects
Mr. Do Quang
TCH03 · quang.do@school.edu
2 subjects
Students
Subjects
Teachers
Enrollments
S06 Teacher Assign Subjects
9:41
‹ Teachers
Teacher
Save
Dr. Hoang Minh
TCH01 · hoang.minh@school.edu
ASSIGNED SUBJECTS (tap to toggle)
Mobile Programming
SUB101
Databases
SUB102
Software Engineering
SUB103
Web Development
SUB104
Save assignments
Students
Subjects
Teachers
Enrollments
S07 Enrollment List
9:41
Enrollments
Nguyen Van An · ST001
SUB101 Mobile Prog. · 2026–2027
8.5
Nguyen Van An · ST001
SUB102 Databases · 2026–2027
—
Tran Thi Binh · ST002
SUB101 Mobile Prog. · 2026–2027
7.0
Le Minh Chau · ST003
SUB103 Soft. Eng. · 2026–2027
9.0
Pham Quoc Dung · ST004
SUB104 Web Dev. · 2026–2027
—
Students
Subjects
Teachers
Enrollments
S08 Enrollment Form
9:41
Cancel
Enroll Student
STUDENT
ST001 – Nguyen Van An
SUBJECT
SUB103 – Software Engineering (3 cr)
ACADEMIC YEAR
2026–2027
Enroll
Cancel



Figma file name: GeoProgramming_Task4_SESEIU23001 Figma link (view access for instructor): _____ (to be added after importing the frames)